

Title:

Thermal and Mechanical Performance of Ceramic Substrates for High-Temperature Power Electronics

Abstract

This paper investigates the suitability of ceramic-based substrates for power electronic modules operating at temperatures above 200°C. Alumina (Al_2O_3), aluminum nitride (AlN), and silicon nitride (Si_3N_4) substrates are evaluated in terms of thermal conductivity, mechanical strength, and long-term reliability.

1. Introduction

The increasing demand for high-power-density electronics in automotive and aerospace applications requires substrates capable of operating under extreme thermal and mechanical stress. Ceramic substrates offer superior thermal stability compared to conventional FR-4 materials.

2. Materials and Methods

Three ceramic materials were evaluated:

- Alumina (Al_2O_3)
- Aluminum Nitride (AlN)
- Silicon Nitride (Si_3N_4)

Samples were metallized using direct bonded copper (DBC) and tested under cyclic thermal loading between -40°C and 250°C.

3. Results

Aluminum Nitride demonstrated the highest thermal conductivity (170–180 W/mK) but showed increased brittleness under mechanical shock. Silicon Nitride exhibited superior fracture toughness and long-term stability under thermal cycling.

4. Discussion

The results indicate that ceramic substrate selection must balance thermal performance with mechanical robustness. For automotive traction inverters, Silicon Nitride substrates offer a favorable compromise.

5. Conclusion

Ceramic-based PCBs represent a viable solution for high-temperature power electronics. Further work will investigate multilayer ceramic interconnects and embedded cooling channels.

Keywords:

Ceramic substrates, power electronics, thermal management, high-temperature PCB